R15

Code No: 124CW

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year II Semester Examinations, December - 2018
ENVIRONMENTAL STUDIES
(Common to CE, ECE, CSE, EIE, IT, MCT, ETM, AME, PTM, CEE)

Time: 3 Hours **Max. Marks: 75**

Note: This question paper contains two parts A and B.
Part A is compulsory which carries 25 marks. Answer all questions in Part A.
Part B consists of 5 Units. Answer any one full question from each unit.
Each question carries 10 marks and may have a, b, c as sub questions.

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PART- A

(25 Marks)

1.a) How does the ecosystem function? [2]
b) What is bioaccumulation? Mention its demerits. [3]
c) Define human development index. [2]
d) What are the sources of pollution for surface water and ground water? Suggest few ways to reduce the pollutant levels. [3]
e) What is meant by genetic diversity? [2]
f) Define hotspots of bio-diversity. Give examples. [3]
g) What are the chemicals that cause ozone layer depletion? [2]
h) What are primary and secondary air pollutants? Give examples for each. [3]
i) Suggest the ways to handle and safe disposal of biomedical wastes from hospital. [2]
j) What is meant by ecological footprint? Mention its value for an individual at the global level. [3]

PART-B

(50 Marks)

2. Describe the role of water cycle and carbon cycle in affecting the energy flow in the ecosystem. [10]

OR

3. Explain the various components of terrestrial food pyramid and aquatic food pyramid. [10]

4.a) Discuss the environmental impacts of building the water reservoirs and dams.
b) Discuss the significant factors that cause flooding and drought. [5+5]

OR

5. Explain the principle involved in the generation of electricity using solar energy. What are the drawbacks of solar energy? [10]

6. Discuss the consumptive use value and productive use value of forest ecosystem in bio-diversity. [10]

OR

7. Explain the reasons for habitat loss and man-wildlife conflicts within the ecosystem. Suggest the ways to reduce the effect. [10]

8. Describe the source, ill effects and control measures for the following classes of water pollutants: a) disease causing agents and b) oxygen depleting agents. [10]

OR

9. Identify the sources, characteristics and environmental impacts of e-waste. Suggest the measures to be taken against the detrimental effects of electronic wastes. [10]

10.a) Explain the role of individuals to prevent domestic level water pollution.
b) Discuss the regulations and guidelines framed in the water pollution control act. [5+5]

OR

11.a) Enumerate the factors that contribute for population explosion.
b) Describe the methods adopted to ensure low carbon life style in our day-to-day activities. [5+5]

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